Document Identifier: **PMC8451553**

SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IRB: All animal experiments were performed in accordance with the Guidelines of the Beijing Animal Ethics Association, and the Ethics Committee of Beijing Institute of Technology (approval No. SYXK-BIT-school of life science-2017-M03) approved the study in 2017.
Inclusion and Exclusion Criteria
not detected.
Attrition
not detected.
Sex as a biological variable
not detected.
Subject Demographics
Age: Twenty-two Tta+Tau+ mice (40 weeks old) were randomly selected from among the hybridized offspring and randomly separated into two groups: NSC-transplanted (Tg + NSCs n = 11) and phosphate-buffered saline (PBS)-treated (Tg + PBS n = 11) groups.
Randomization
Twenty-two Tta+Tau+ mice (40 weeks old) were randomly selected from among the hybridized offspring and randomly separated into two groups: NSC-transplanted (Tg + NSCs n = 11) and phosphate-buffered saline (PBS)-treated (Tg + PBS n = 11) groups.
Blinding
not detected.
Power Analysis
not detected.
Replication
Number: The brain slices were rinsed with 50 ethanol , 20 ethanol , and pure water sequentially , with each rinse step performed three times for 10 minutes each .

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
After blocking with 5 horse serum in PBST for 30 minutes , the NSCs were incubated with the following primary antibodies: bIII tubulin (a specific marker for neurons , mouse monoclonal , Santa Cruz Biotechnology , Santa Cruz , CA , USA , Cat sc-80005 , 1:400) glial fibrillary acidic protein (GFAP , a specific marker for astrocytes , goat polyclonal , Abcam , Cambridge , UK , Cat ab53554 , 1:500) and ionized calcium-binding adapter molecule 1 (Iba1 , a specific marker for microglia , goat polyclonal , Abcam , Cat ab5076 , 1:500) .	bIII tubulin		
	GFAP		
	ionized calcium-binding adapter molecule 1		
	Iba1		
Primary antibodies were incubated at 4degC for 20 hours , and then the cells were washed three times with PBS for 5 minutes , after which the NSCs were incubated with the following secondary antibodies at room temperature for 2 hours: Alexa Fluor 488 labeled anti-mouse IgG (Beyotime , Shanghai , China , A0428 , 1:500	A0428		
The NSCs were then incubated with anti-GFP antibody (chicken polyclonal , Abcam , Cat 13970 , 1:500 , 0.3 prepared in PBST) at 4degC for 20 hours , followed by Alexa Fluor 488 labeled anti-chicken IgG (Beyotime , Cat A0428 , 1:500) secondary antibody at room temperature for 2 hours .	anti-chicken IgG		
The brain slices were incubated with the chicken-derived GFP monoclonal primary antibody (chicken polyclonal , Abcam , Cat ab13970 , 1:1000 , prepared in 3 horse serum in PBST) at 4degC for 16 hours .	GFP		
The brain slices were then rinsed three times in PBS for 5 minutes each and incubated with a second primary antibody , either neuronal neuron (NeuN , mouse monoclonal , Abcam , Cat ab104224 , 1:400) , glial fibrillary acidic protein (GFAP , goat polyclonal , Abcam , Cat ab53554 , 1:500) or Iba1 (goat polyclonal ,	NeuN		
	glial fibrillary acidic protein		

Abcam , Cat ab5076 , 1:500) at 4degC for 20 hours .			
The secondary antibodies were diluted in PBST as follows: donkey anti-goat IgG H&L (Alexa Fluor(r) 555 Abcam , ab150130 , 1:500)	anti-goat IgG		
	ab150130		
The PVDF membrane was blocked in 2 bovine serum albumin for 2 hours at room temperature and then incubated overnight at 4degC with the following primary antibodies: extracellular regulated protein kinase 12 (ERK12 rabbit , Cell Signaling , Danvers , MA , USA , Cat 4695 , 1:2000) , phosphorylated-ERK12 (rabbit , Cell Signaling , Cat 9101 , 1:2000) , cyclic adenosine monophosphate-response element-binding protein (CREB rabbit , Cell Signaling , Cat 4034 , 1:2000)	ERK12		
	phosphorylated-ERK12		
	CREB		
The PVDF membrane was then incubated with the secondary antibody (horseradish peroxidase-labeled goat anti-mouse IgG , 1:5000) at room temperature for 2 hours .	anti-mouse IgG		
Experimental Models: Organisms/Strains			
Materials and Methods Animals For this study, rTg (tau P301L) 4510 mice (referred to as Tg-tau mice) were obtained through the hybridization between transgenic tau mice (FVB, gifted by the Xuanwu Hospital in Beijing, China) and the transgenic Tta mice (C57BL6, purchased from the Jackson Laboratory, Bar Harbor, ME, USA).	FVB		
Extraction of primary NSCs Primary NSCs were extracted from the hippocampus of C57BL6 wild-type (WT) suckling mice on postnatal day 1 (purchased from the Department of Laboratory Animal Science, Peking University Medical School, Beijing, China, license No. SCXK (Jing) 2016-0010).	C57BL6		Suggestion: RRID:IMSR_JAX:000664(link)
Software and Algorithms			
, phosphorylated CREB (rabbit , Cell Signaling , Cat 4095 , 1:2000) , nuclear factor kappa beta (NFkB rabbit , Cell Signaling , Cat 3034 , 1:2000) , phosphorylated-NFkB (rabbit , Cell Signaling , Cat 3034 , 1:2000) , tau (rabbit ,	Thermos Fisher Scientific		

1:1000 , Abcam , Cat ab47579) , phosphorylated-paired helical filaments (PHF)-tau pSer202+Thr205 (AT8) (mouse , Thermos Fisher Scientific , Waltham , MA , USA , Cat MN1020 , 1:1000) , Ca2+calmodulin-dependent			
The protein bands were visualized using an enhanced chemiluminescence detection system (Bio-Rad , Hercules , CA , USA) , and bands were measured and analyzed using ImageJ software (National Institutes of Health , Bethesda , MD , USA) .	ImageJ		Suggestion: ImageJ, RRID:SCR_003070(link)
(KEGG) pathway analysis (https:www.genome.jpkegg) and Gene Ontology (GO) classification (http:geneontology.org).	KEGG		
The protein interactions were analyzed by STRING software (http: https:string-db.org).	STRING		

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
T-maze test The T-maze test was used to examine the spatial working memory abilities of mice (Deacon and Rawlins , 2006) .	T-maze test
After Students t-test , differentially expressed proteins were filtered according to P < 0.05 and Fold-Change > 1.5 .	t-test
Statistical analysis Comparisons between two groups were performed using unpaired, two-tailed Students t-test, and multiple comparisons were performed by one-way analysis of variance followed by Tukeys post hoc tests.	analysis of variance
	Tukeys post hoc tests

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq